

Module 04: Cognition -- The Models Inside Your Head

Learning Objectives

1. Describe a **generative model** as the brain's internal representation of the causal structure of the world, including beliefs, expectations, and assumed relationships.
2. Explain how cognitive biases emerge naturally from efficient but sometimes inaccurate generative models.
3. Analyze the difference between **model updating** (changing beliefs to fit evidence) and **active inference** (changing the world to fit beliefs).
4. Evaluate the role of emotions as signals about the state of your generative model -- indicators of prediction error, uncertainty, and model confidence.

Introduction

You carry a model of the world inside your head. It is not a perfect copy -- it is a simplified, compressed representation of the patterns your brain considers most important. This model predicts what will happen when you walk into a room, what a friend means by a certain look, what grade you will get on a test, and what kind of future awaits you. It runs continuously, mostly below conscious awareness, shaping your thoughts, feelings, and decisions.

In Active Inference, this internal model is called a **generative model** because it *generates* predictions about sensory experience. Cognition, in this framework, is the process of maintaining, running, and updating this model. Understanding your own generative model is perhaps the most powerful form of self-knowledge available.

Key Concepts

1. What Is a Generative Model?

A generative model is your brain's theory of how the world works -- specifically, a theory of how *causes* in the world produce the *effects* you observe. It includes beliefs about physical laws (dropped objects fall), social dynamics (people who smile are friendly), and personal

patterns (I perform well under pressure). The model is "generative" because it can produce simulated sensory data: you can imagine what a conversation will look like before it happens, which means your model is generating predicted experience.

2. Beliefs as Model Parameters

In Active Inference, a **belief** is not an opinion you consciously hold -- it is a parameter in your generative model that shapes predictions. Many beliefs operate below awareness. You believe that solid floors will support your weight (which is why you do not test the floor before each step). You believe that your name, when called, refers to you. These implicit beliefs are the backbone of your generative model, and they are only noticed when they produce prediction errors -- when the floor gives way or someone uses your name to address someone else.

3. Cognitive Biases as Model Features

Cognitive biases -- confirmation bias, anchoring, availability heuristic -- are often taught as errors in thinking. Active Inference reframes them as *features* of an efficient generative model. Confirmation bias (seeking evidence that supports existing beliefs) is a consequence of the fact that your model assigns high prior probability to its own predictions. Anchoring (over-relying on the first piece of information) is a consequence of strong priors dominating weak evidence. These are not bugs; they are how an efficient inference engine operates under limited time and energy. But they *can* lead to persistent errors.

4. Emotions as Model Diagnostics

Why do you feel anxious before a big test? Why does rejection sting? Active Inference suggests that emotions are **signals about the state of your generative model**. Anxiety signals high expected prediction error -- your model is uncertain about what will happen and predicts potential threats. Sadness signals a significant model update -- something important in your model (a relationship, an expectation about the future) has been proven wrong. Joy signals low prediction error in domains you care about -- things are going as predicted or better. Emotions are not irrational disruptions; they are informative readouts from your inference machinery.

5. The Depth of the Model

Your generative model is not flat; it is **hierarchical**. Surface-level predictions (what the next word in this sentence will be) are generated by mid-level models (what this paragraph is about), which are generated by high-level models (what this module is trying to teach). Similarly, your prediction about how a conversation will go is nested within your model of the relationship, which is nested within your model of social dynamics in general. Deeper levels of the model change more slowly and have wider-reaching consequences when they do change.

Active Inference Connection

Cognition, in the Active Inference framework, is the process of optimizing internal states (beliefs) to minimize variational free energy. This optimization has two components: **perceptual inference** (updating beliefs to better explain current evidence) and **active inference** (selecting actions to bring the world into alignment with existing beliefs). The balance between these two strategies is central to mental life. Excessive perceptual inference means changing your beliefs at every challenge, leading to instability. Excessive active inference means rigidly trying to make the world match your expectations, leading to inflexibility. Healthy cognition balances both.

Applications

- **Case Study 1 -- Why Changing Your Mind Feels Uncomfortable:** When you encounter strong evidence against a deeply held belief -- a political view, a conviction about a friend's character, a career aspiration -- the resulting prediction error propagates up through your hierarchical model. Updating a deep belief is costly because it ripples through many dependent predictions. This is why people often resist changing their minds even in the face of clear evidence: the free energy cost of a deep model revision can be enormous. Understanding this through Active Inference does not make it easier, but it makes the discomfort intelligible.
- **Case Study 2 -- Rumination and Anxiety Loops:** Rumination -- going over the same worrying thoughts repeatedly -- can be understood as a generative model that is stuck in

a local minimum. The model predicts threat, this prediction generates anxiety, and the anxiety increases precision weighting on threat-related information, which confirms the prediction. The model is minimizing free energy locally (its predictions match the anxiety-filtered evidence it attends to) but failing globally (the overall level of free energy remains high). Breaking out of this loop often requires external intervention that introduces new evidence the model cannot dismiss.

Discussion Questions

1. Identify a belief you hold that operates below conscious awareness -- something so basic you never question it. What would happen if it turned out to be wrong?
2. If emotions are signals about the state of your generative model, how might understanding this change the way you respond to feelings of anxiety or sadness?
3. Why might it be more difficult to change a deep-level belief (like your political identity) than a surface-level one (like your opinion of a movie)?

Summary

Cognition is the maintenance and operation of your generative model -- the internal representation of reality that drives your predictions, shapes your perceptions, and gives rise to your emotions. Cognitive biases are features of efficient inference, not character flaws, though they can lead to persistent errors. Emotions serve as diagnostic signals about model states. Understanding your own cognition through Active Inference gives you a vocabulary for self-reflection that is both precise and deeply practical. In Module 05, we turn from internal models to external behavior: **action** -- how your generative model drives the decisions you make.